

PRAKASH

AI Engineer

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PROFESSIONAL SUMMARY

Final year B.Tech AI/ML student (CGPA 8.07) with hands-on experience in building production-ready AI systems using PyTorch, LangChain, and large language models. Developed end-to-end projects in computer vision, generative AI, and Agentic RAG systems. Passionate about solving real-world problems and seeking opportunities as an AI/ML Engineer to build scalable and intelligent solutions.

TECHNICAL SKILLS

Languages	Python, SQL (MySQL), HTML, CSS
ML / DL	PyTorch, TensorFlow, Keras, Scikit-learn, Neural Networks, Transfer Learning, Transformer Architecture, OpenCV
Gen AI / LLMs	LangChain, Ollama, RAG, Prompt Engineering, LLM Fine-tuning, AI Agents, Multi-Agent Systems, ChromaDB, Vector DBs
Frameworks	Flask, FastAPI, Pandas, NumPy, Matplotlib
MLOps & Tools	Git, GitHub, Docker, Kubernetes, CI/CD Pipelines, REST APIs, Model Deployment, API Integration

PROJECTS

Agentic RAG Document Assistant <i>Python • Pandas • Flask • LLMs</i>	GitHub ↗ Jan 2026
- Built an end-to-end Agentic RAG data analytics platform enabling conversational AI-driven data analysis through a full-stack web application using Flask, Pandas, and Groq LLMs. - Developed an intelligent agentic routing system that automatically classified user queries and executed dynamic data-filtering workflows with structured JSON-based parameter extraction. - Solved reliable query-routing and response-accuracy challenges by implementing prompt-engineered function calling, modular AI workflows, and context-aware data processing pipelines.	
AI-Powered Cattle Breed Identification <i>Python • PyTorch • ResNet18 • FastAPI • Google Gemini AI</i>	GitHub ↗ Jul 2025
- Built an AI-powered cattle breed identification system for real-time image-based breed classification using deep learning and FastAPI. - Trained a ResNet18 transfer-learning model on 23,000+ images across 86 breeds with data augmentation for improved accuracy. - Solved visually similar breed-classification challenges by integrating confidence-based predictions and Gemini AI-powered breed insights.	
Cook Craft AI — Intelligent Recipe Generator <i>Python • Flask • Ollama • Gemma 2B • HTML/CSS</i>	GitHub ↗ Nov 2025
- Built an offline AI-powered recipe generator that created personalized recipes from user ingredients, cuisine preferences, and cooking constraints using Gemma 2B via Ollama. - Developed a Flask-based web application with dynamic prompt engineering to generate structured step-by-step cooking instructions in real time without external APIs. - Solved fast and reliable offline recipe-generation challenges by optimizing prompt workflows and designing a responsive, user-friendly recipe visualization interface.	

EDUCATION

Sri Shakthi Institute of Engineering and Technology	Coimbatore, Tamil Nadu
<i>B.Tech — Artificial Intelligence and Machine Learning CGPA: 8.07 / 10</i>	Sep 2023 – Present
Model School	Salem, Tamil Nadu
<i>Higher Secondary (Class XII) Percentage: 80%</i>	Jun 2022 – Jun 2023

CERTIFICATIONS

Cloud Computing	- NPTEL — IIT Kharagpur	2024
Artificial Intelligence Fundamentals	- IBM	2026